0.1 基本组合学公式

定义 0.1 (组合数定义的扩充)

对 $\forall n \in \mathbb{R}, k \in \mathbb{N}$, 定义

$$\mathrm{C}_n^k = \binom{n}{k} \triangleq \begin{cases} 0 & , k < 0, \\ \dfrac{n(n-1)\cdots(n-k+1)}{k!} & , 0 < k \leqslant n, \\ 0 & , k > n. \end{cases}$$

特别地,$\mathrm{C}_n^0 \triangleq 1$. 若 $n, k \in \mathbb{N}$ 且 $0 \leqslant k \leqslant n$, 则还有

$$\mathrm{C}_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

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命题 0.1 (组合数基本公式)

1. 对 $\forall n, m \in \mathbb{N}$, 有 $\mathrm{C}_n^m = \mathrm{C}_{n-1}^m + \mathrm{C}_{n-1}^{m-1}$.
2.

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证明 Show/Hide Proof

□

命题 0.2

$$\sum_{i=0}^{m} (-1)^i \binom{n}{i} = (-1)^m \binom{n-1}{m}, \quad 0 \leqslant m \leqslant n-1.$$

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证明 Show/Hide Proof

$$\begin{aligned}
\sum_{i=0}^{m} (-1)^i \binom{n}{i} &= \sum_{i=0}^{m} (-1)^i \left[\binom{n-1}{i} + \binom{n-1}{i-1}\right] = \sum_{i=0}^{m} (-1)^i \binom{n-1}{i} + \sum_{i=0}^{m} (-1)^i \binom{n-1}{i-1} \\
&= \sum_{i=0}^{m} (-1)^i \binom{n-1}{i} + \sum_{i=0}^{m-1} (-1)^{i+1} \binom{n-1}{i} = \sum_{i=0}^{m} (-1)^i \binom{n-1}{i} - \sum_{i=0}^{m-1} (-1)^i \binom{n-1}{i} \\
&= (-1)^m \binom{n-1}{m}.
\end{aligned}$$

□

定理 0.1 (二项式定理的推广)

$$\begin{aligned}
\prod_{i=1}^{n} (a_i + b_i) &= \sum_{S \subset \{1,2,\cdots,n\}} \left(\prod_{i \in S} a_i \prod_{j \in \{1,2,\cdots,n\} \setminus S} b_j \right) \\
&= \prod_{i=1}^{n} a_i + \sum_{m=1}^{n} \sum_{\substack{|S|=m, \\ S \subseteq \{1,2,\cdots,n\}}} \left(\prod_{i \in S} a_i \prod_{j \notin S} b_j \right) \\
&= \prod_{i=1}^{n} b_i + \sum_{m=1}^{n} \sum_{\substack{|S|=m, \\ S \subseteq \{1,2,\cdots,n\}}} \left(\prod_{i \in S} b_i \prod_{j \notin S} a_j \right).
\end{aligned}$$

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证明 Show/Hide Proof

利用数学归纳法证明即可.

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